

Euler Product and Riemann Zeta Function Formalizations

Lean Formalization Project

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This document provides a comprehensive list of all definitions manually formalized for the analytic proof of the Prime Number Theorem via Newman's Tauberian approach, specifically within `Contour.lean` and `Zeta.lean`. For each definition, the raw Lean code is presented first, followed by its mathematical interpretation.

1 ComplexAnalysis/Sarason/Contour.lean

- `newmanModifier` ($s \in \mathbb{C}$, $R \in \mathbb{R}$): The modifying factor in Newman's contour integral to guarantee convergence on the left half-plane.

```
noncomputable def newmanModifier (s : ℂ) (R : ℝ) : ℂ :=  
  (1 : ℂ) + s^2 / (R^2 : ℂ)
```

$$\text{newmanModifier}(s, R) = 1 + \frac{s^2}{R^2}$$

- `newmanModifier_eq_on_circle`: Equality bounding the modifier on the circle $|s| = R$.

```
lemma newmanModifier_eq_on_circle (s : ℂ) (R : ℝ) (hR : R > 0) (hs : Complex.  
  abs s = R) :  
  newmanModifier s R = (s + R^2 / s) / s
```

$$|s| = R \implies 1 + \frac{s^2}{R^2} = \frac{s + R^2/s}{s}$$

2 ComplexAnalysis/Sarason/Zeta.lean

- `zetaPartialSum` ($s \in \mathbb{C}$, $N \in \mathbb{N}$): Partial sum of the Riemann zeta function up to integer N .

```
noncomputable def zetaPartialSum (s : ℂ) (N : ℕ) : ℂ :=  
  ∑ i ∈ Finset.range N, if i = 0 then 0 else ((i : ℂ) ^ (-s))
```

$$\zeta_N(s) = \sum_{n=1}^{N-1} n^{-s}$$

- `eulerPartialProduct` ($s \in \mathbb{C}$, $N \in \mathbb{N}$): Partial Euler product over primes up to N .

```
noncomputable def eulerPartialProduct (s : ℂ) (N : ℕ) : ℂ :=  
  ∏ p ∈ (Finset.range N).filter Nat.Prime, (1 - ((p : ℂ) ^ (-s)))^-1
```

$$P_N(s) = \prod_{p < N} \frac{1}{1 - p^{-s}}$$

- `IsEulerProductAt_eps / eulerProduct_eps`: The absolute limit of the Euler product as $N \rightarrow \infty$.

```
def IsEulerProductAt_eps (s : ℂ) (L : ℂ) : Prop :=
  ∀ ε > 0, ∃ N₀, ∀ N ≥ N₀, Complex.abs (eulerPartialProduct s N - L) < ε

noncomputable def eulerProduct_eps (s : ℂ) : ℂ :=
  if h : ∃ L, IsEulerProductAt_eps s L then Classical.choose h else 0
```

$$P(s) = \lim_{N \rightarrow \infty} P_N(s)$$

- `zeta_eq_eulerProduct`: Euler product identity for $\text{Re}(s) > 1$.

```
lemma zeta_eq_eulerProduct (s : ℂ) (hs : s.re > 1) :
  zeta_eps s = eulerProduct_eps s
```

$$\text{Re}(s) > 1 \implies \zeta(s) = \prod_p \frac{1}{1 - p^{-s}}$$

- `geom_sum_inv_expansion`: Geometric series expansion identity.

```
lemma geom_sum_inv_expansion (x : ℂ) (hx : Complex.abs x < 1) (K : ℕ) :
  (1 - x)⁻¹ = (∑ k ∈ Finset.range K, x^k) + (x^K) * (1 - x)⁻¹
```

$$\frac{1}{1 - x} = \sum_{k=0}^{K-1} x^k + \frac{x^K}{1 - x} \quad \text{for } |x| < 1$$

- `prime_euler_factor_inv`: Expansion specifically for prime factors.

```
lemma prime_euler_factor_inv (p : ℕ) (s : ℂ) (hs : s.re > 0) (K : ℕ) (hp :
  Nat.Prime p) :
  (1 - ((p : ℂ) ^ (-s)))⁻¹ = (∑ k ∈ Finset.range K, ((p : ℂ) ^ (-s))^k) +
    (((p : ℂ) ^ (-s))^K) * (1 - ((p : ℂ) ^ (-s)))⁻¹
```

$$\frac{1}{1 - p^{-s}} = \sum_{k=0}^{K-1} (p^{-s})^k + \frac{p^{-sK}}{1 - p^{-s}}$$

- `SmoothNumbers` ($N \in \mathbb{N}$): Integers with prime factors bounded by N .

```
def SmoothNumbers (N : ℕ) : Set ℕ :=
  {n : ℕ | n > 0 ∧ ∀ p, p.Prime → p | n → p ≤ N}
```

$$\text{SmoothNumbers}(N) = \{n \in \mathbb{N}^+ \mid p \mid n \implies p \leq N\}$$

- `smooth_numbers_fta_bijection`: The Fundamental Theorem of Arithmetic (FTA) Bijection Lemma for Smooth Numbers.

```
lemma smooth_numbers_fta_bijection (N : ℕ) (n : ℕ) (hn : n ∈ SmoothNumbers N)
  :
  ∃! (f : ℕ →₀ ℕ), (∀ p ∈ f.support, p.Prime ∧ p ≤ N) ∧ f.prod (fun p k =>
    p ^ k) = n
```

Every integer in $S(N)$ has a unique prime factorization over $p \leq N$.

- `smoothNumbersBoundedFinset` ($N, K \in \mathbb{N}$):

```

noncomputable def smoothNumbersBoundedFinset (N K : ℕ) : Finset ℕ :=
  have : DecidablePred (fun n => n > 0 ∧ ∀ p, p.Prime → p | n → p ≤ N ∧ n.
    factorization p ≤ K) := Classical.decPred _
  (Finset.range ((N.factorial ^ K) + 1)).filter (fun n => n > 0 ∧ ∀ p, p.Prime
    → p | n → p ≤ N ∧ n.factorization p ≤ K)

```

$$\text{SmoothNumbersBounded}(N, K) = \{n \leq (N!)^K \mid p \mid n \implies p \leq N \text{ and } v_p(n) \leq K\}$$

- `euler_product_expand_sum`: The polynomial expansion of the Euler product into the geometric zeta sum.

```

lemma euler_product_expand_sum (N K : ℕ) (s : ℂ) :
  ∏ p ∈ (Finset.range (N + 1)).filter Nat.Prime, (∑ k ∈ Finset.range (K +
    1), ((p : ℂ) ^ (-s)) ^ k) =
  ∑ n ∈ smoothNumbersBoundedFinset N K, (n : ℂ) ^ (-s)

```

$$\prod_{p \leq N} \left(\sum_{k=0}^K p^{-sk} \right) = \sum_{n \in S(N, K)} n^{-s}$$